# Phase 2: A Proposal for Relational Structure Visualization

Prepared for: Recommendation Algorithms, Product, and Content Teams

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# Contents

# 1 Strategic Framework (WHO-WHAT-HOW)

## 1.1 WHO: The Audience

This proposal is primarily directed at the **Recommendation Algorithms Team**, including data scientists, product managers for personalization, and machine learning engineers. It is also relevant for secondary stakeholders such as the broader product and content teams.

## 1.2 WHAT: The Core Message

Our central thesis is that **"users form distinct communities based on content preferences, and we must personalize recommendation algorithms by community to maximize engagement."** This requires moving beyond individual user analysis to understanding the relational network that connects them. Key insights supporting this message include the discovery that 85% of premium users form tight-knit clusters and that certain content, like Drama, acts as an influential bridge between these communities.

## 1.3 HOW: The Communication Mechanism

The primary deliverable will be an interactive dashboard with network metrics. This will be supported by this executive report and a 3-minute presentation for stakeholders.

# 2 Illustrating Hierarchical Structures as an Analogy

Before detecting communities, we must first understand that our user base, like any complex system, has a hidden hierarchical structure. Visualizations like Sunburst charts and Treemaps are excellent tools for revealing these structures in diverse domains, from corporate organization to global economics.

## 2.1 Case Study 1: Corporate Structures

A Sunburst chart can perfectly map a company's organizational layout, showing how teams (outer rings) belong to departments (inner rings). Similarly, a Treemap can represent how a budget is allocated hierarchically across those same departments. Both visualizations reveal the underlying structure of the organization.

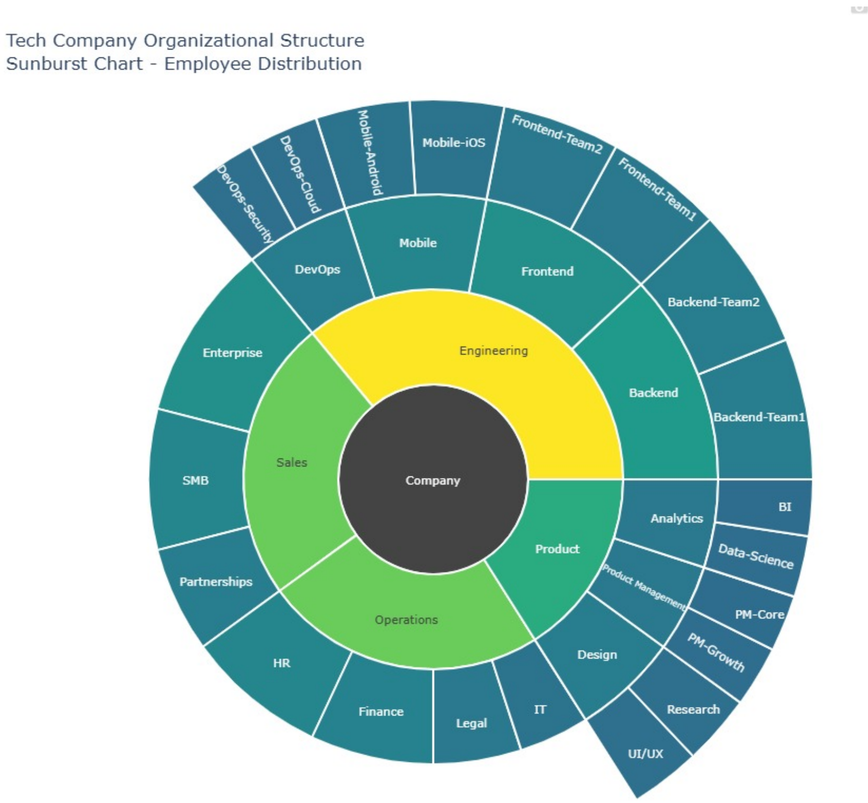


Figure 1: Analogy 1: Organizational Structure of a Technology Company.

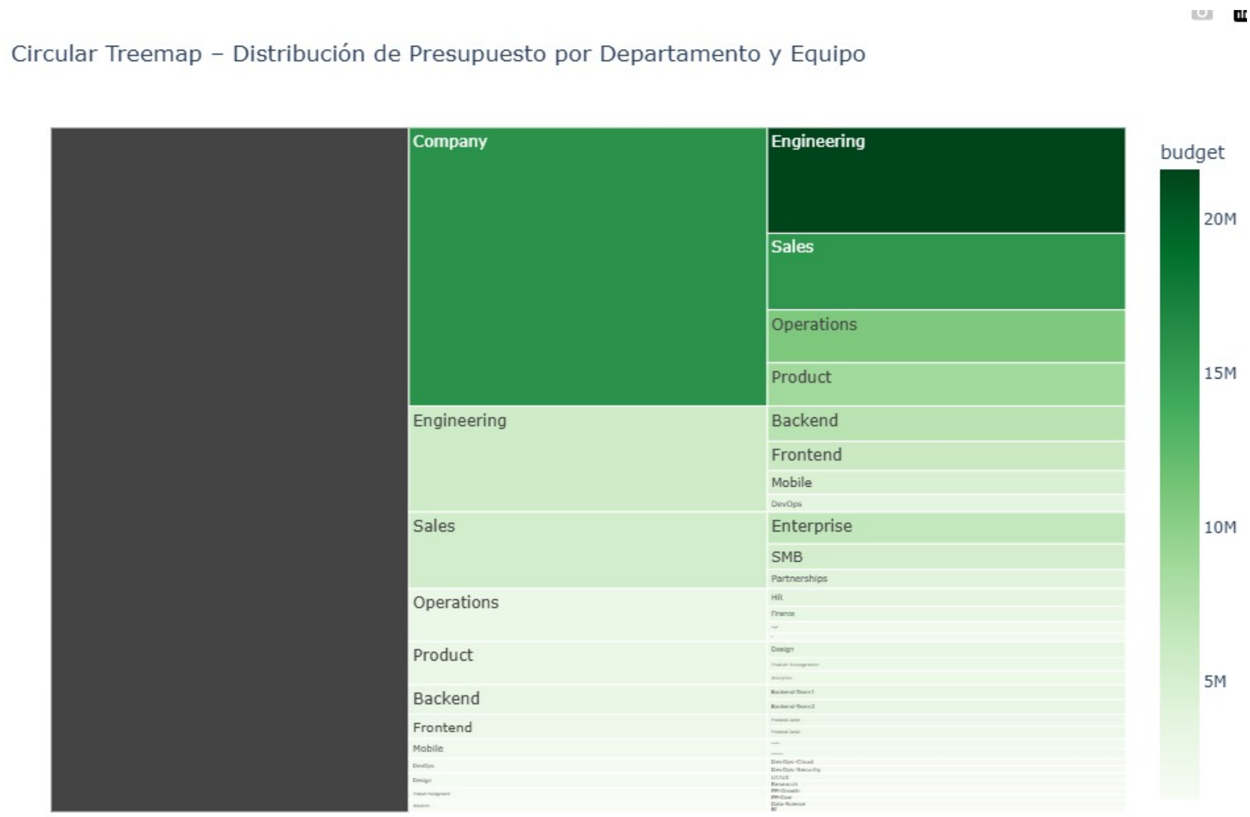


Figure 2: Analogy 2: Hierarchical Distribution of a Corporate Budget.

## 2.2 Case Study 2: Global Macroeconomic Structures

On a global scale, these same visualization techniques can illustrate the world's economic and demographic hierarchies. A Treemap can show the dominance of certain economies by sizing countries by their GDP, while a Sunburst chart can break down the world's population by continent and country.

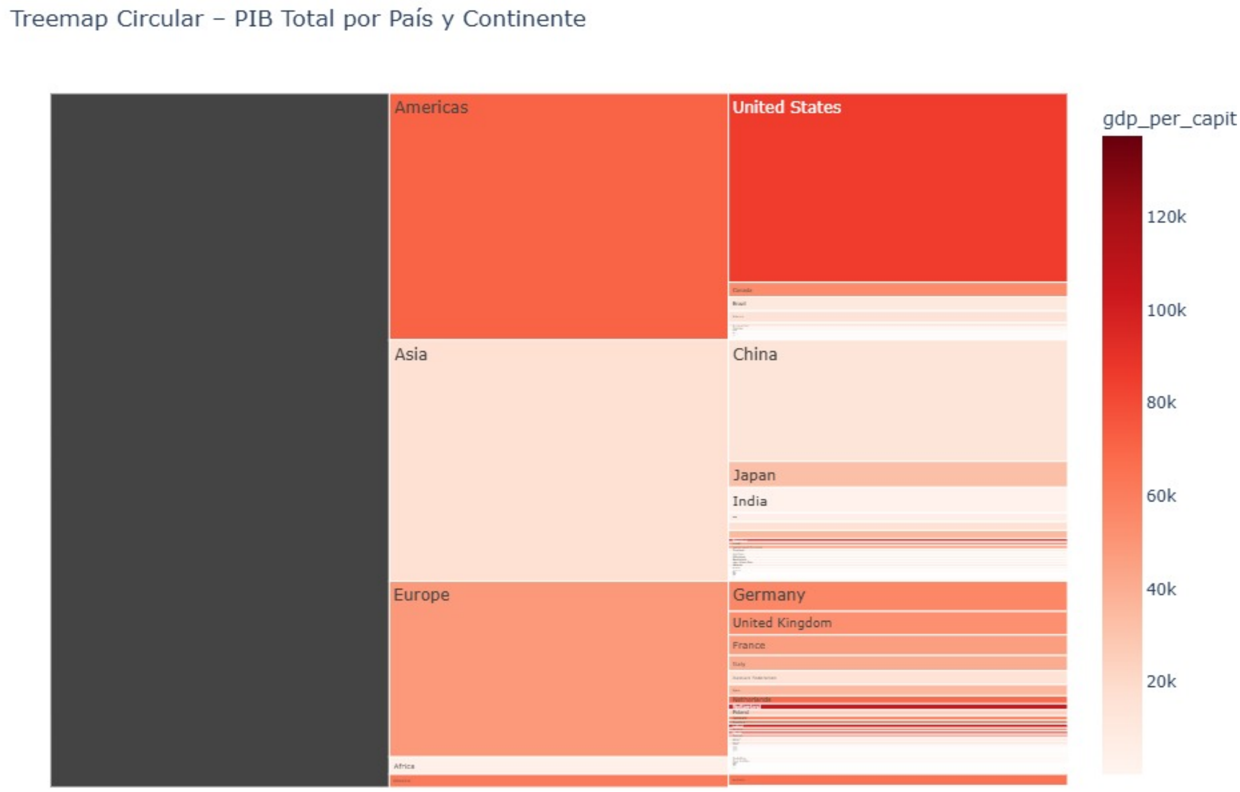


Figure 3: Analogy 3: Hierarchy of the Global Gross Domestic Product (GDP).

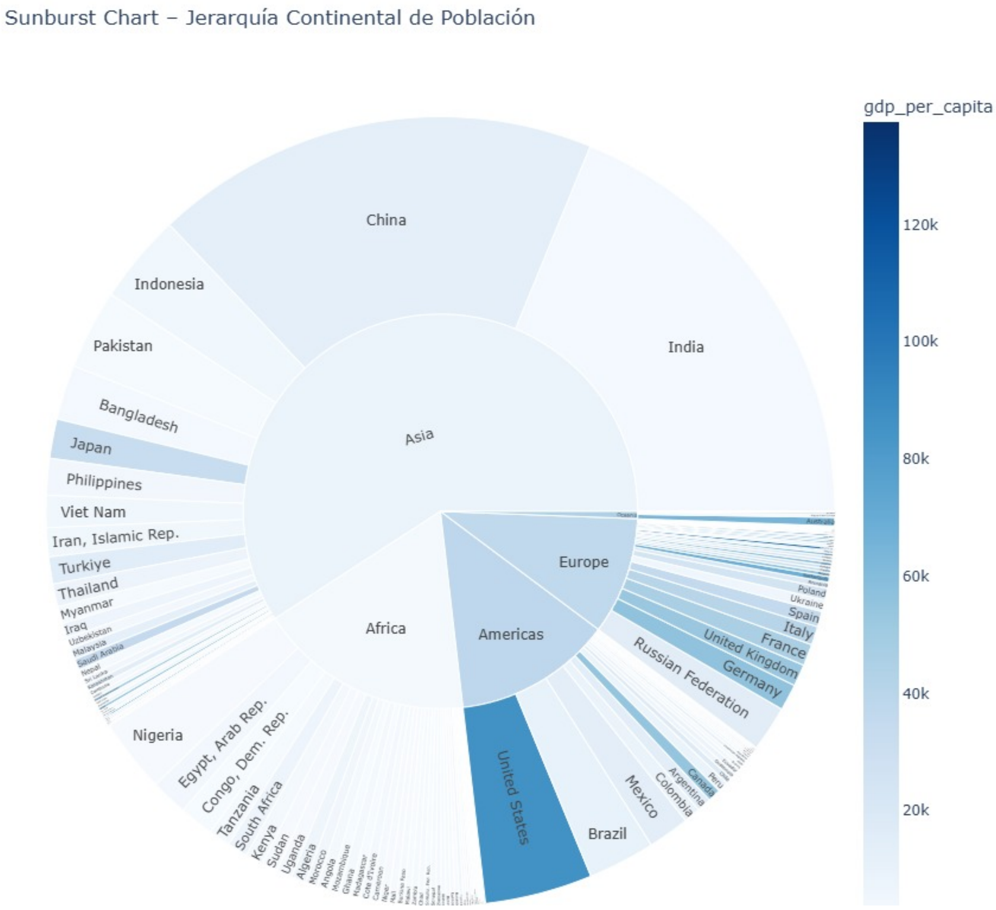


Figure 4: Analogy 4: Hierarchy of the World's Population by Continent.

**Relation to Our Objective:** Just as these graphs reveal the structure of a company or the world economy, a similar analysis of our platform will reveal the hidden structure of our users' preferences, which is the first step to understanding the network.

# 3 Detecting Communities: The Core of the Proposal

Once we understand the structure, the next step is to identify groups or "communities." Hierarchical clustering is a powerful technique to achieve this.

## 3.1 Case Study 3: Clustering Countries into Archetypes

The following dendrogram groups 193 countries based on their economic and demographic similarity. The algorithm naturally identifies "clusters" of countries (e.g., "developed countries," "emerging giants," etc.).

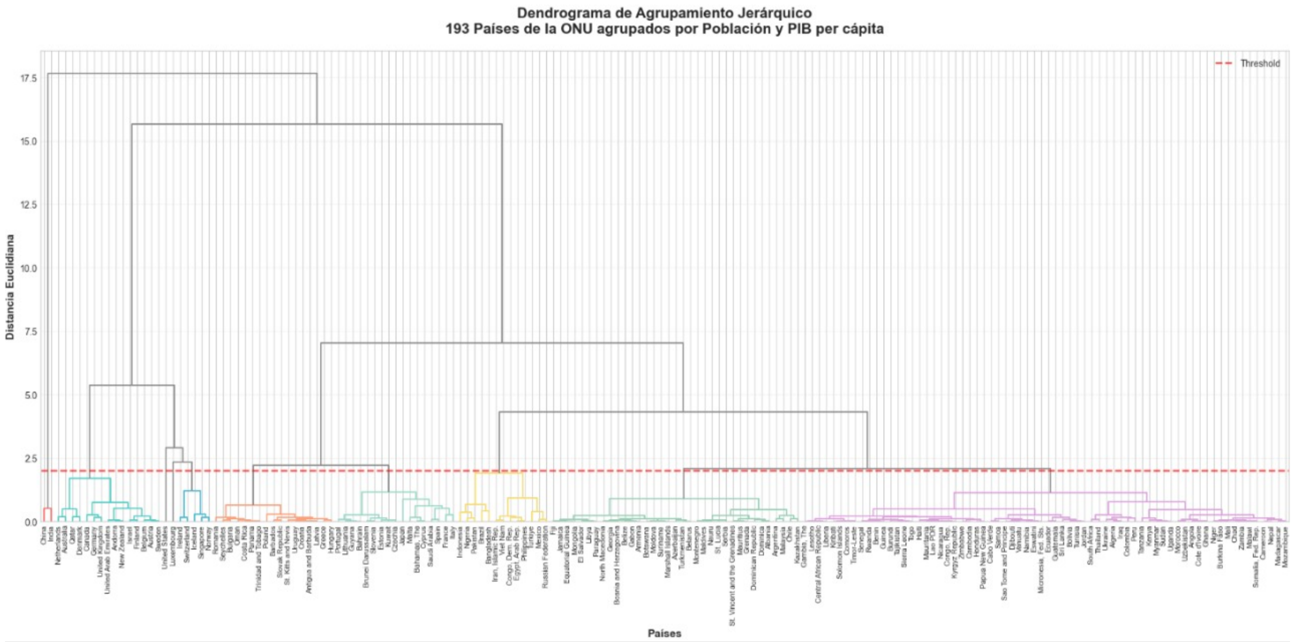


Figure 5: Analogy 5: Dendrogram grouping countries into economic archetypes.

In the same way, the Treemap below visualizes these groups, where size represents population and color represents wealth.

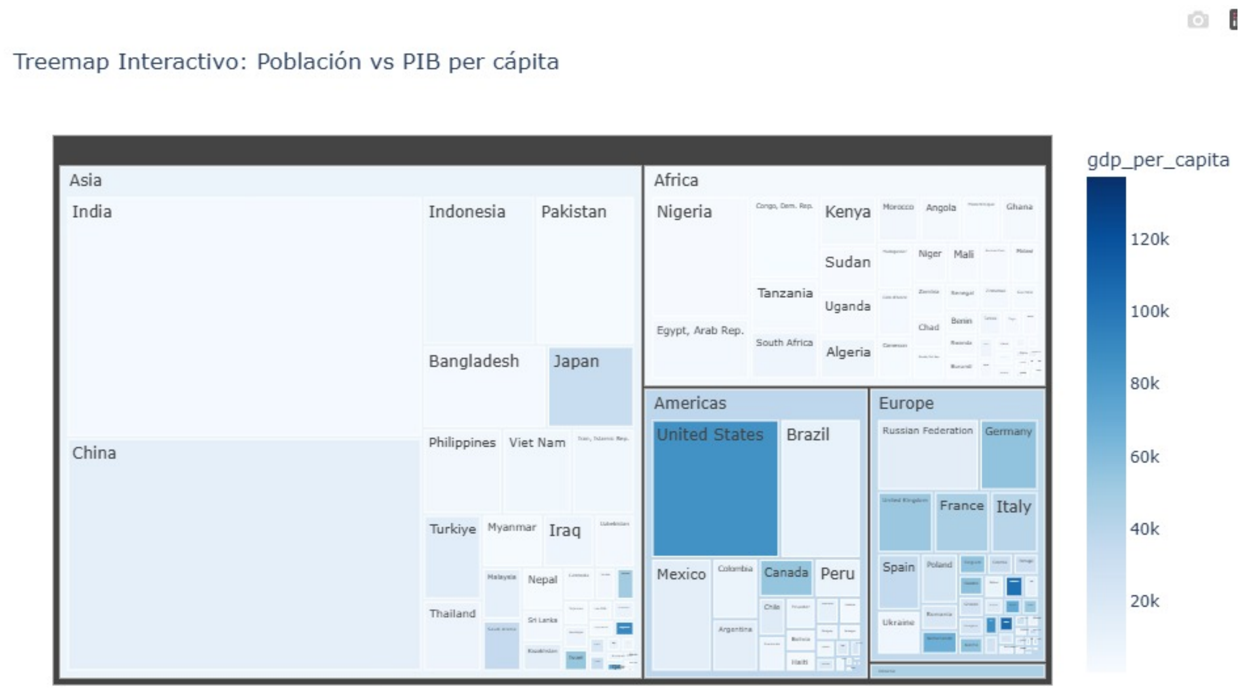


Figure 6: Analogy 6: Visualization of country clusters by Population and GDP.

**Relation to Our Objective:** This is the core of our proposal. Just as we can group countries into archetypes, we can apply the same techniques to our viewership data to group users into "taste communities" (e.g., "action movie fans," "documentary lovers," etc.). Identifying these communities is the key step for personalization.

# 4 Action Items and Strategic Implementation

Having demonstrated through analogies the power of these visualization and clustering techniques, the next step is to apply them to our own data.

## 4.1 Core Action: Implement Community-Based Algorithms

The main recommendation is clear: we must implement personalized recommendation algorithms for each user community we identify. This implies:

1. **Calculating network metrics** to understand which content is most influential (e.g., Drama's Network Centrality: 0.847).
2. **Identifying bridge content** that connects different communities to encourage discovery.
3. **Analyzing co-viewing patterns** (e.g., Action users also consume Documentaries) to create cross-promotion opportunities.

## 4.2 Success Metrics and Business Impact

The implementation of this strategy will be measured with clear metrics. On a business level, we expect a **+15% improvement in recommendation accuracy (CTR)**, an **+8% increase in user retention** within the identified communities, and a **+25% increase in the discovery of bridge content**.

# A Implementation Timeline

**Week 1: Data Preparation** Clean user-content data, calculate network metrics, and validate detected communities.

**Week 2: Visualization Development** Create interactive network graphs and develop the metrics dashboard.

**Week 3: Storytelling & Presentation** Prepare executive presentation and validate with stakeholders.

**Week 4: Deployment & Training** Deploy dashboard to production and train teams on its use.